Incident investigation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which observation supports the hypothesis and what test could challenge it.

Teach prompt containment, privacy-correct evidence and bounded causal claims.

Keep one habit: Model timeline inference → contrast retrieval/draft mismatches → inspect controlled fixture → assess permission-config replay plan → record gaps.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of incident investigation

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Monitoring quality drift. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of monitoring quality drift

Reference example, not the only acceptable wording: Teacher explanation: Monitoring compares current outcomes with a declared reference and sampling protocol.

A change in observed quality may reflect case mix or instrumentation. For each window report eligible counts, passes, missing outcomes and condition mix.

Alert when success falls below the toy **80%** threshold; investigate before attributing cause.

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of deployment environments

Reference example, not the only acceptable wording: Teacher explanation: Development, staging and production are distinct environments with explicit data and credential boundaries.

Test configuration must not silently target production. For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names.

A staging run cannot use production write credentials.

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of data retention and deletion

Reference example, not the only acceptable wording: Teacher explanation: Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Pictured end goal

Explain incident investigation using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Monitoring quality drift

Deployment environments

Data retention and deletion

Use the three-session teaching cycle above, with extra time as needed. Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of incident investigation

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach observations, hypotheses, containment and tests as distinct records

> Do not turn a timeline or a mock reproduction into a proven live root cause.

Definitions and complete fictional evidence contract

Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**.

Evidence or action	Complete classroom meaning
Observation	Supplied timestamp, version, trace field or outcome; preserve source and uncertainty
Hypothesis	Proposed explanation that must be compared with evidence
Main trusted version record	Weight version w7 before and at response; c2 published at 09:00, response uses c1 at 09:05
Containment	Cache disabled at 09:06 in the supplied incident record; limits exposure, does not establish cause
Candidate controlled test	Same query/source/weight fixture; vary only whether buggy query-only cache is used
Permitted investigation record	Fictional IDs, times, versions, actions and necessary outcome summaries; no unnecessary private content
Picture evidence localization	Old retrieval version versus current retrieval followed by wrong draft are different mismatches
Limits	Reliable declared metadata assumed; no live reproduction, complete timeline guarantee or causal proof from temporal order

Main timeline ledger and checked reasoning

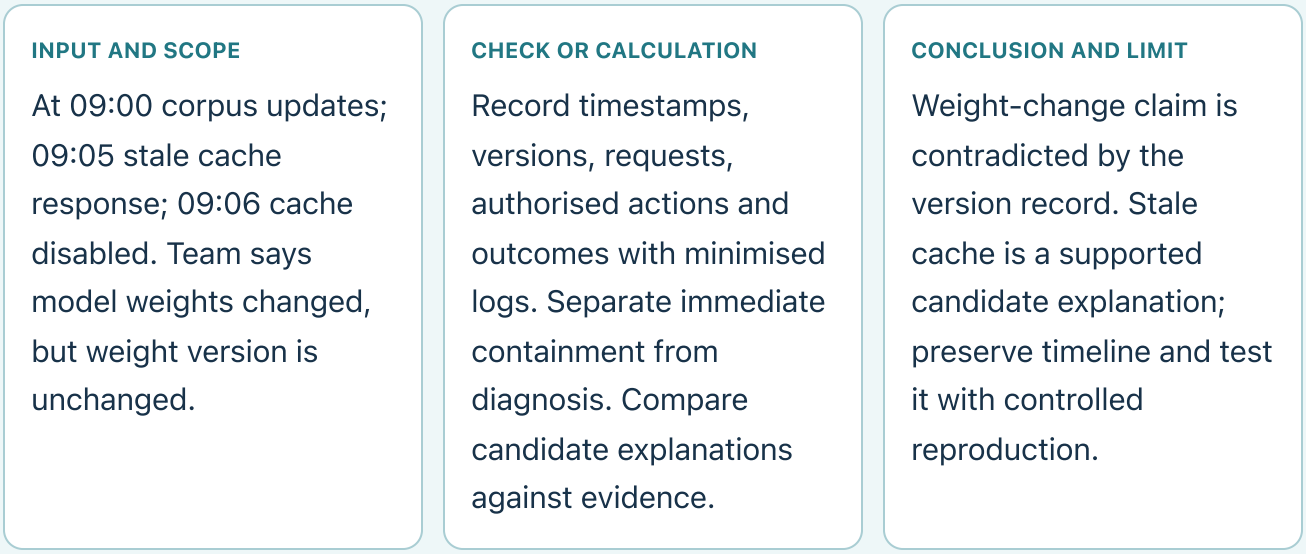
Main supplied event	Observation	What it supports and does not prove
Before response	Weight version w7	Baseline under reliable supplied version metadata
09:00	Corpus c2 published	New source predates the stale response
09:05	Response uses old c1; weight version still w7	Stale evidence observed; reported weight-change explanation conflicts with declared metadata
09:06	Cache disabled	Containment action; no supplied later success rate
Added controlled fixture	Cache enabled gives c1; disabled gives c2	A buggy query-only cache can reproduce this pattern locally; exact real bug remains unproven

Main reliable metadata reports unchanged w7, so a claimed weight-version change conflicts with the supplied record. Old c1 response after c2 publication supports stale evidence as a candidate; chronology does not name the exact bug. The controlled deliberately buggy cache reproduces old/current fixture selection but is not a live diagnosis.

Picture afternoon events use version 5 room C after version 6 room D publishes; changed version 6 retrieval plus room C draft instead points to handoff/drafting/checking. Capture authorized minimized evidence while containing promptly; do not teach delaying needed containment or keeping unnecessary private content.

Selected permission-config incident needs an old/new configuration replay of allowed and denied fictional cases, with other settings fixed and expected access outcomes declared.

Incident investigation — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged.

Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized **evidence** without unnecessary private content so a later revision can be compared honestly.

Evidence, privacy and containment probes

Investigation probe	Expected interpretation
Event B follows event A	Temporal order only; not sufficient causal proof
Trusted declared weight version unchanged	Contradicts a reported weight-version change under that metadata assumption
Old source version appears in trace	Evidence-selection mismatch; test version routing and cache candidates
Current source retrieved, incompatible room drafted	Inspect handoff/drafting/validation; do not label it old retrieval
Disabling cache recorded	Containment, not proof the incident is fully resolved
Candidate reproduced in a fictional harness	Supports possible mechanism in that harness; compare actual permitted incident evidence
Missing version or incomplete events	Mark evidence gap; do not invent it
Necessary containment is urgent	Proceed promptly while collecting authorized minimized evidence where feasible
Private payload unnecessary to analysis	Exclude it; preserve permitted metadata instead

Separate afternoon timeline and changed trace

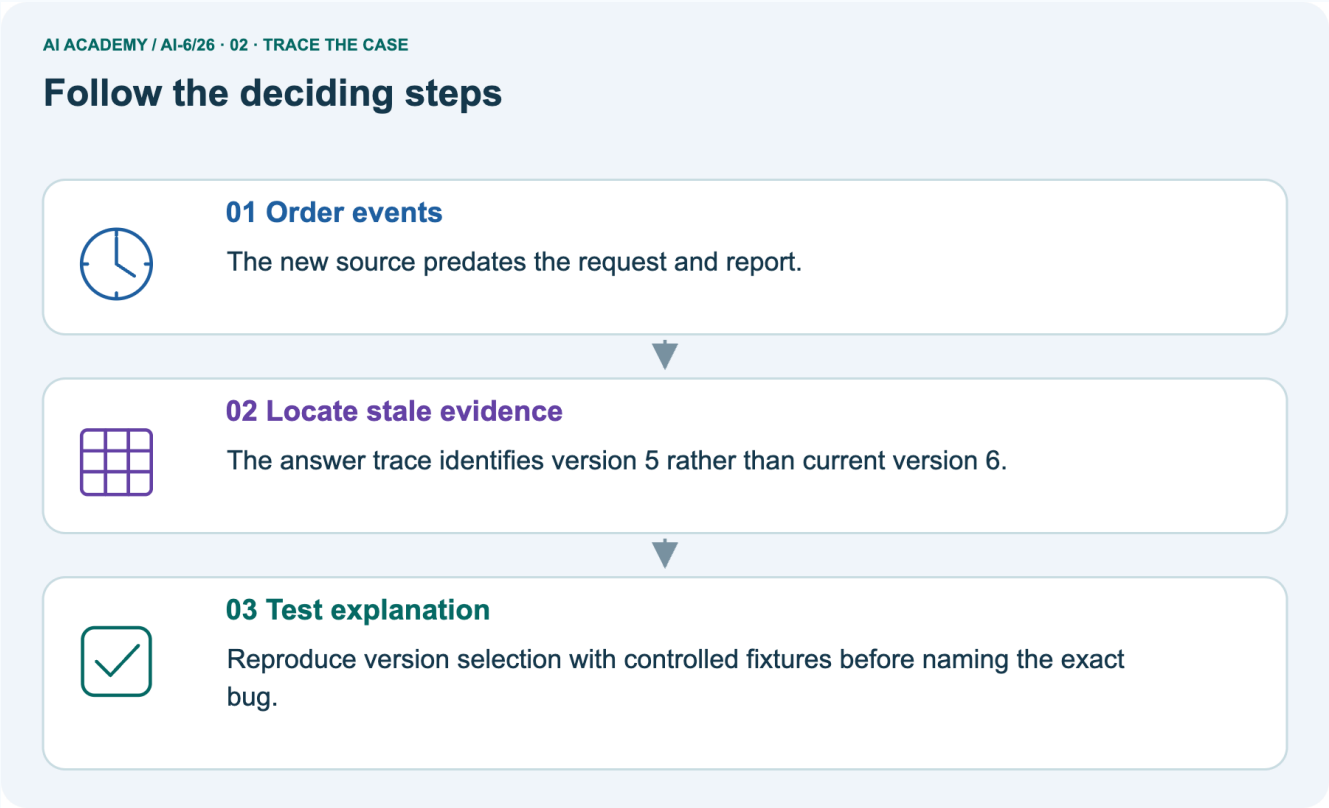
An **incident timeline**. After a schedule update, a service gives an old classroom location.

14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

The picture is a separate afternoon incident: 14:00 version 6 publishes room D; 14:05 the trace uses version 5 and drafts room C; 14:06 a user reports the mismatch. This locates stale evidence selection, but does not identify the exact code defect. Controlled version-selection/cache tests are appropriate.

In the changed trace, version 6 was retrieved but room C was drafted; now inspect context handoff, answer construction and output checking rather than asserting old-source retrieval. It is still possible for a cache elsewhere in a system to matter; this trace alone does not diagnose every component. Missing events or unreliable metadata may change the account.

Picture trace and evidence-localization key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Order events	The new source predates the request and report.
2. Locate stale evidence	The answer trace identifies version 5 rather than current version 6.
3. Test explanation	Reproduce version selection with controlled fixtures before naming the exact bug.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A timeline separates observed events from a proposed cause. **Evidence** can locate a mismatch without yet explaining why it happened.

Missing events may change the account; temporal order alone does not prove causation.

Change: Trace instead shows version 6 retrieved and room C drafted.

Prediction: Investigate drafting or validation before blaming caching.

Reason: The mismatch now appears after retrieval.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Incident timeline	Ordered records of relevant events
Containment	An action limiting ongoing harm or failure
Root-cause hypothesis	A proposed mechanism requiring evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

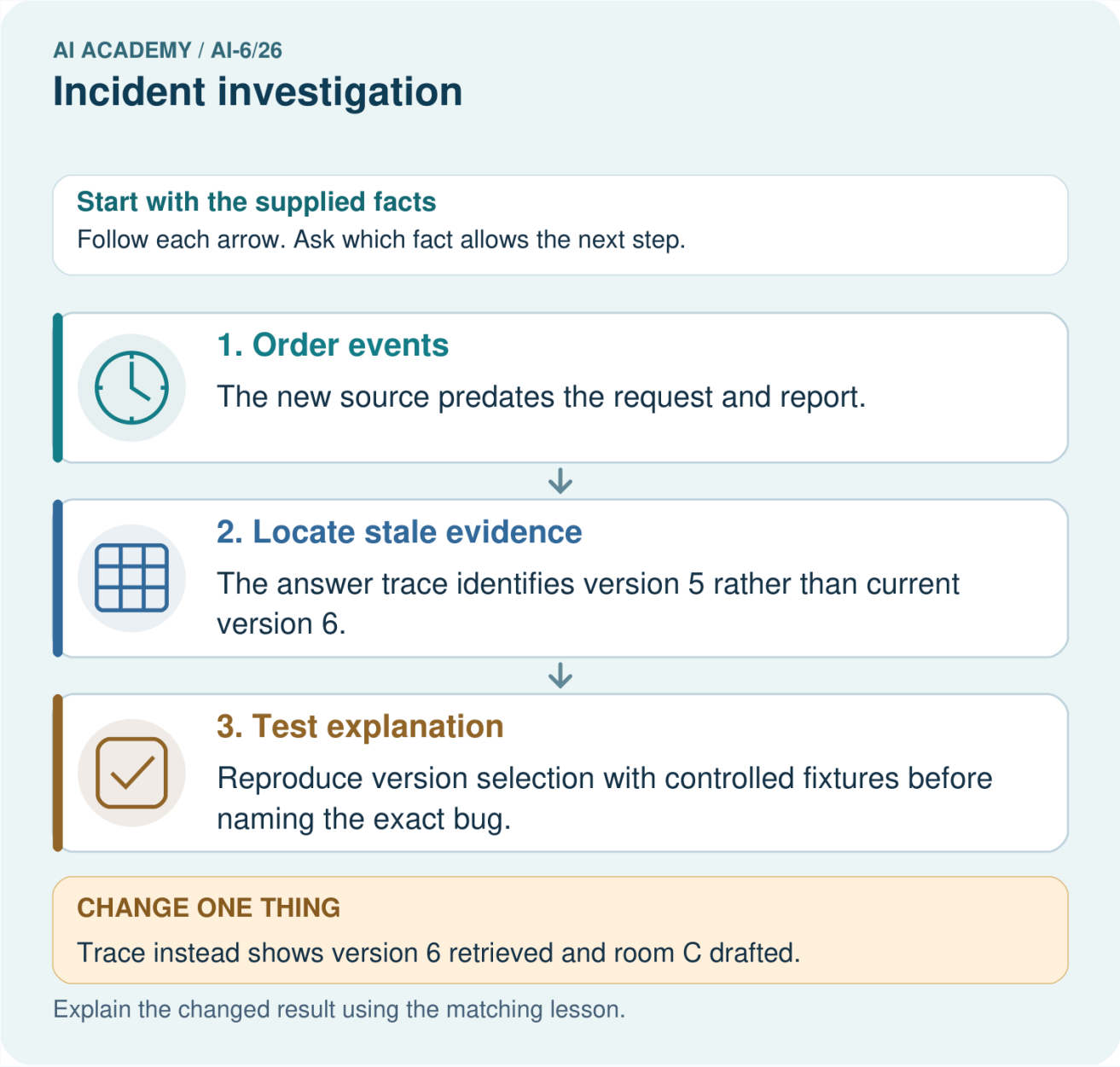
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Incident investigation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Incident investigation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Order the three incident records and locate the old **evidence** version. How does the investigation change if the trace instead retrieves version 6 but drafts room C?

The new source predates the request and report. The answer trace identifies version 5 rather than current version 6. Reproduce version selection with controlled fixtures before naming the exact bug.

Changed case: Investigate drafting or validation before blaming caching. The mismatch now appears after retrieval.

Changed-case reasoning: Investigate drafting or validation before blaming caching. The mismatch now appears after retrieval.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main unchanged declared weight version challenges a weight-change claim; stale cache is a candidate to test, not a proven cause. Picture old version 5 supports evidence-selection investigation; current version 6 with room C draft changes focus to handoff/drafting/checks.

Selected 10:00 permission-config change and 10:03 restricted read: investigate configuration as a plausible candidate; replay authorized and denied fictional cases against old/new configurations with other settings fixed, expected outcomes recorded and no real restricted content exposed. Temporal sequence is insufficient proof.

Diagnose the current misconceptions

The worked result establishes every future case: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: 10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

Teacher explanation: Investigate permission/config change; replay authorised and denied cases against old/new config without exposing real restricted content. Temporal order alone is not proof.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**.

W2 Trace the worked case

At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

W3 Audit the limit

A report omits this qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

Teacher explanation: Investigate permission/config change; replay authorised and denied cases against old/new config without exposing real restricted content. Temporal order alone is not proof.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Investigate permission/config change; replay authorised and denied cases against old/new config without exposing real restricted content. Temporal order alone is not proof.

Require intermediate reasoning and the stated scope. Fresh retry: A queue redelivery follows a crash after commit. Two effects appear.

Expected reasoning: Inspect operation-ID deduplication and durable status; timeline supports investigation, not an assumed complete cause.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled incident investigation **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of incident investigation.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Human escalation workflows. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/27, ai-6/29, ai-6/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of incident investigation

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Check every learner before discussion: Use the worked case for Incident investigation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Trace instead shows version 6 retrieved and room C drafted. Require a **prediction** and a reason before comparing. Expected result: Investigate drafting or validation before blaming caching.

Reason: The mismatch now appears after retrieval.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

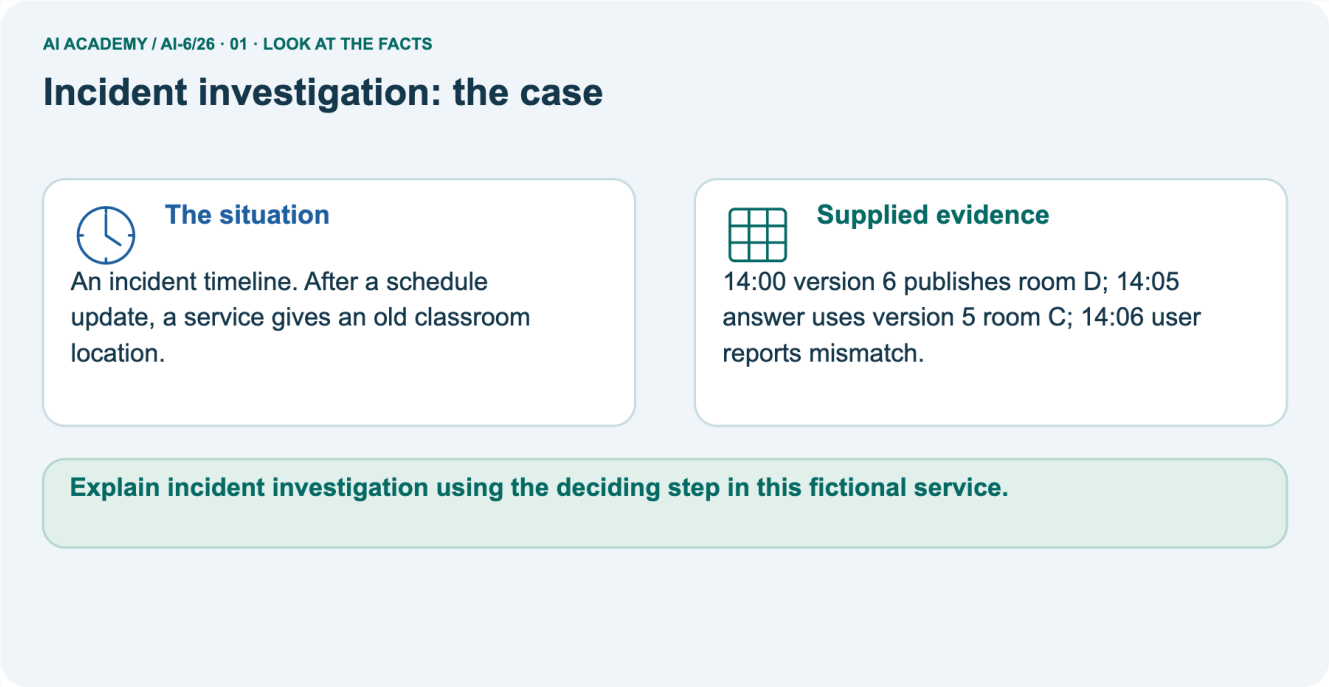
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Incident investigation: the case



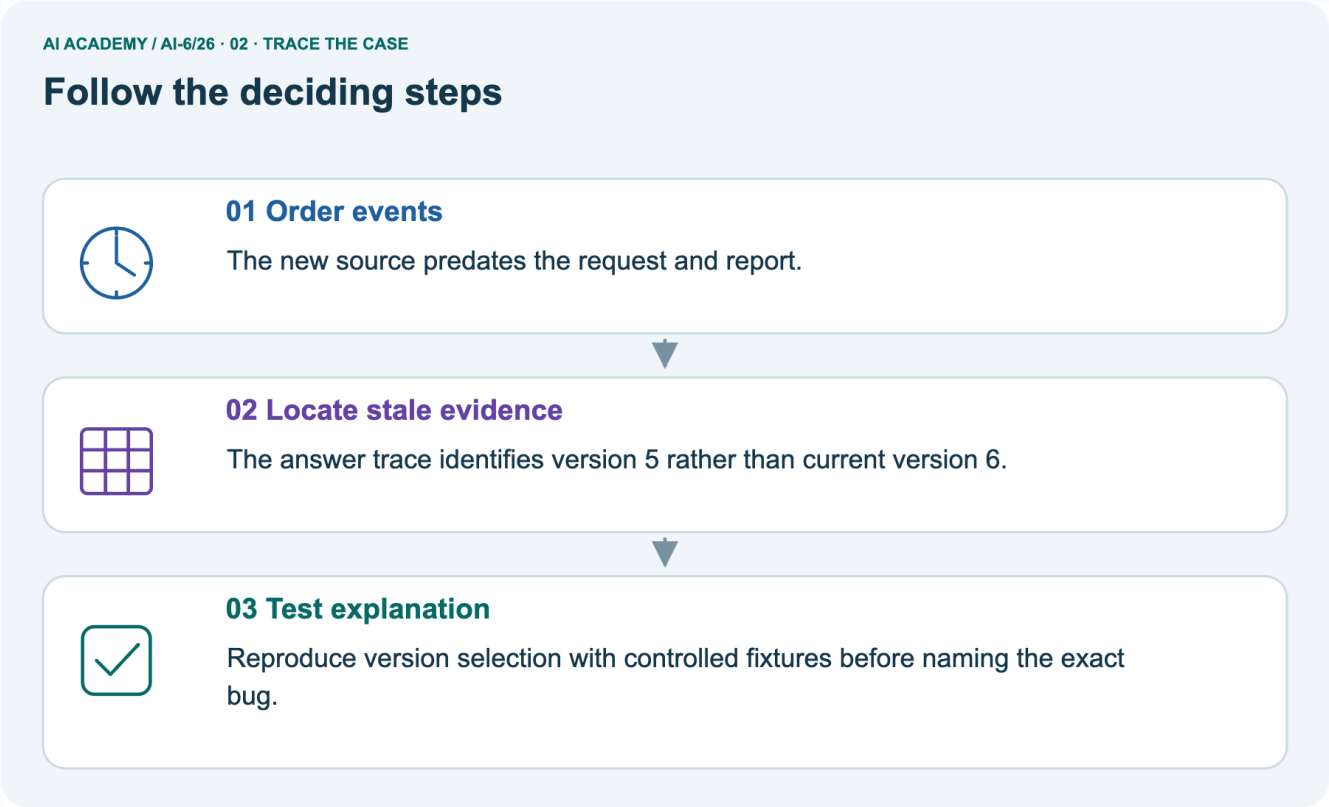
Follow the labelled picture with the matching explanation. Open the vector diagram

An **incident timeline**. After a schedule update, a service gives an old classroom location. Supplied facts: 14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



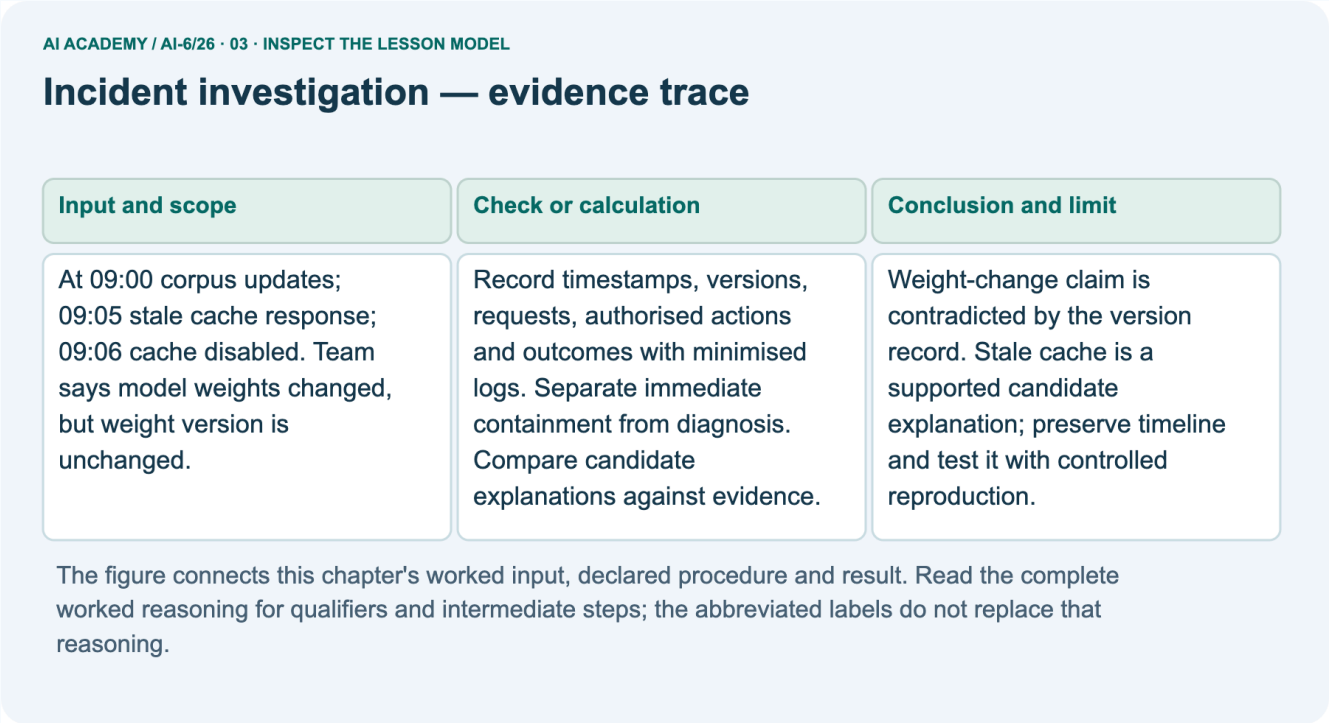
Follow the labelled picture with the matching explanation. Open the vector diagram

Order events: The new source predates the request and report. → Locate stale **evidence**: The answer trace identifies version 5 rather than current version 6. → Test explanation: Reproduce version selection with controlled fixtures before naming the exact bug.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Incident investigation — evidence trace



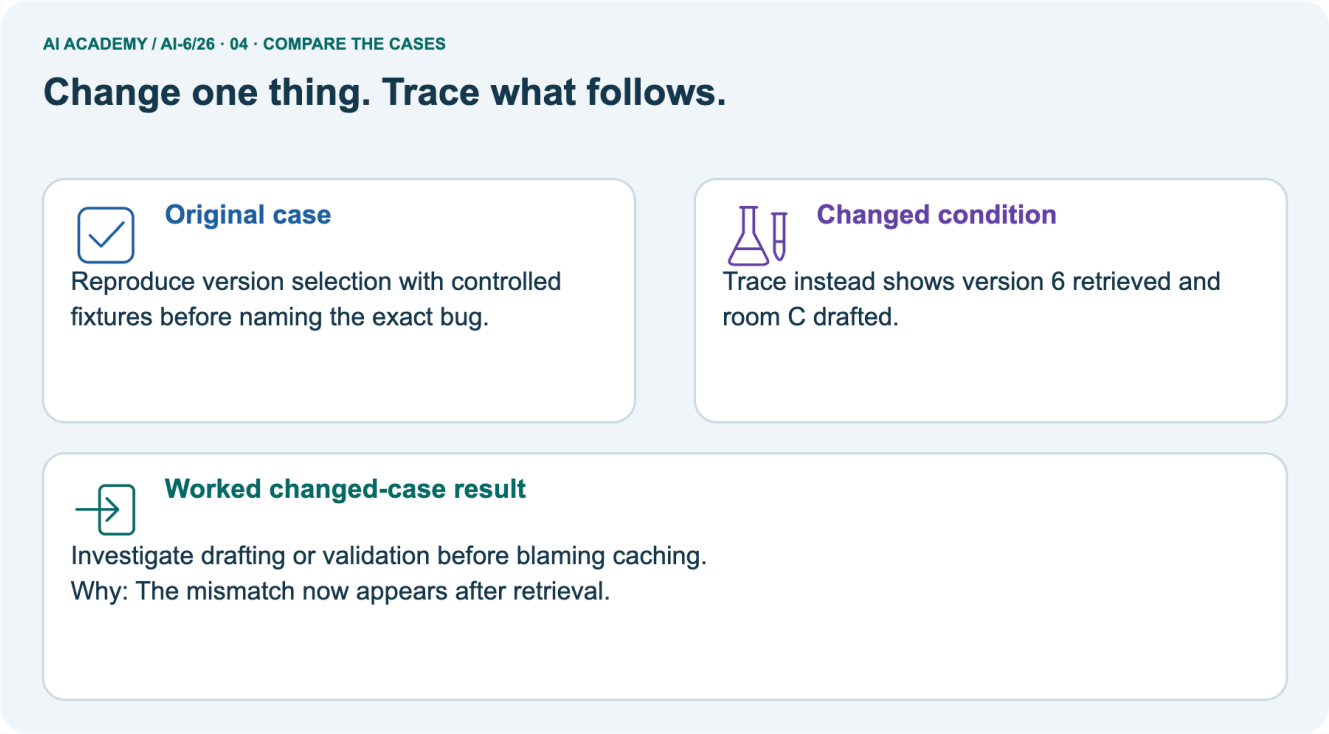
Follow the labelled picture with the matching explanation. Open the vector diagram

Incident investigation — **evidence** trace. At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reproduce version selection with controlled fixtures before naming the exact bug. Changed condition: Trace instead shows version 6 retrieved and room C drafted.

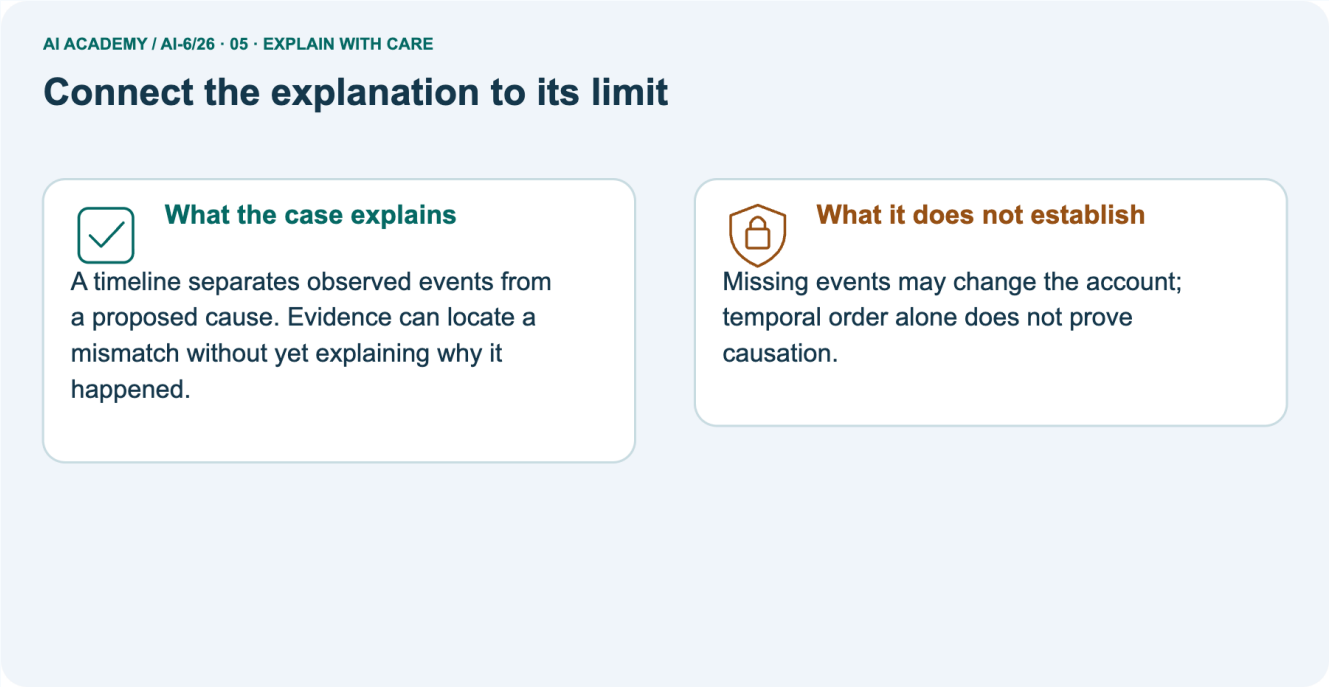
Expected result: Investigate drafting or validation before blaming caching. Reason: The mismatch now appears after retrieval.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The mismatch now appears after retrieval.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

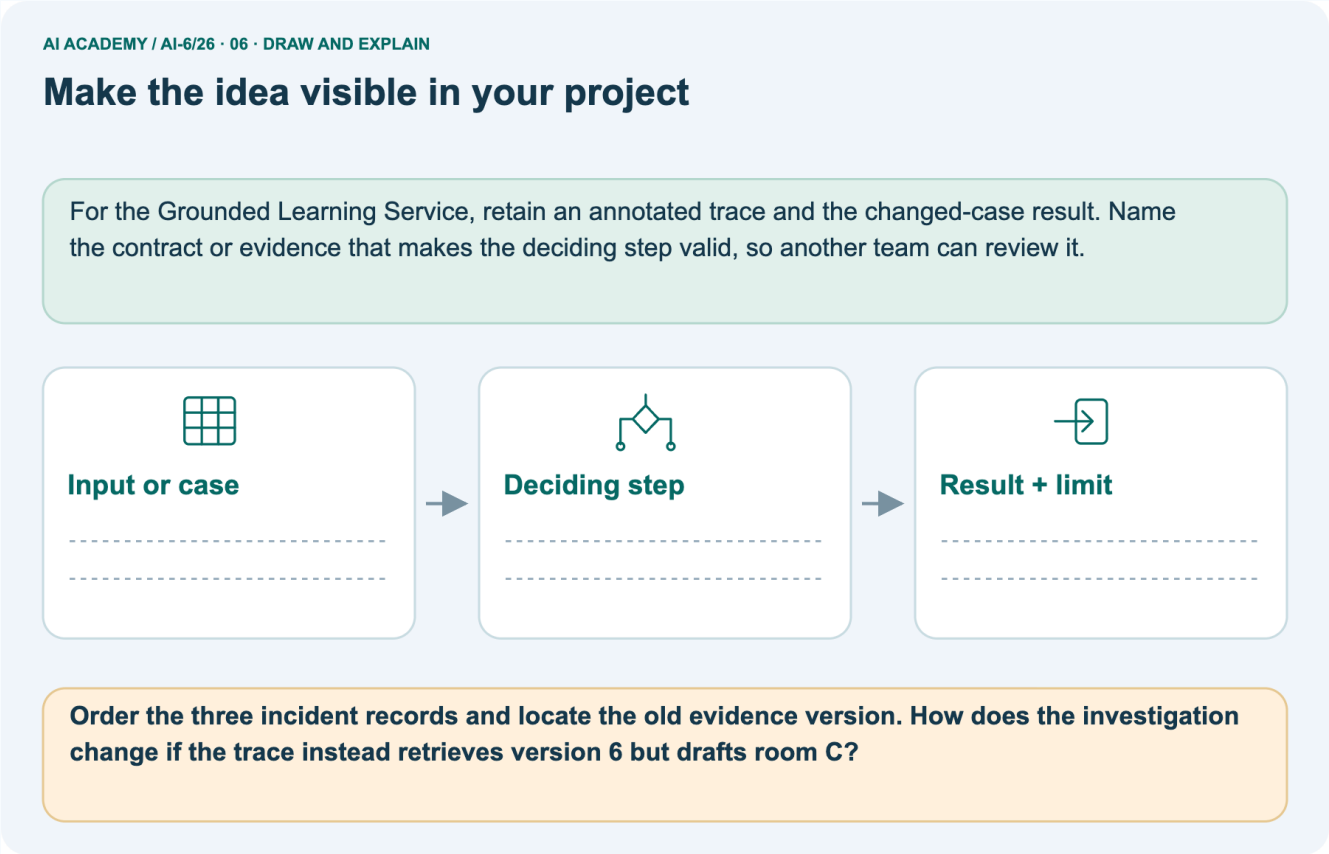
A timeline separates observed events from a proposed cause. **Evidence** can locate a mismatch without yet explaining why it happened. Limit: Missing events may change the account; temporal order alone does not prove causation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Missing events may change the account; temporal order alone does not prove causation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The new source predates the request and report. The answer trace identifies version 5 rather than current version 6. Reproduce version selection with controlled fixtures before naming the exact bug.

Changed case: Investigate drafting or validation before blaming caching. The mismatch now appears after retrieval.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Incident investigation: the case

02. Follow the deciding steps

03. Incident investigation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.